

Generation Interconnection System Impact Study Report

For Queue Project Q22-2167

Yahara River 138 kV

July 17, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2167. The Interconnection Customer proposes a 167.1 MW battery energy storage facility interconnecting at the Yahara River 138 kV substation in the New Jersey area. The request was received on August 26, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 736 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 49 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$51,761,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
COLORADO-NIETOS 138.0 kV	DAM HEIGHTS-BABINE LAKE 138.0 kV	195.4	192	18.5
OLETA-MONTCLAIR COGEN FACILITY 138.0 kV	MULTITRADE OF PITTSYLVANIA LP-SNE - SANSUM N ELEC 138.0 kV	181.2	224	16.9
WELLS RIV SUBSTATION-GASTONIA PRIME POWER PARK 161.0 kV	LEE VINING-CLOSTER 161.0 kV	177.2	280	6.2
SPADRA-MONTCLAIR COGEN FACILITY 138.0 kV	MULTITRADE OF PITTSYLVANIA LP-SNE - SANSUM N ELEC 138.0 kV	160.6	223	16.9
COLORADO-OLETA SUBSTATION 138.0 kV	SNE - SANSUM N ELEC-MULTITRADE OF PITTSYLVANIA LP 138.0 kV line	160.3	281	16.9
ATHENIA-DAM HEIGHTS 138.0 kV	STANDARD OIL 3-BABINE LAKE 138.0 kV	159.4	214	7.4
SADDLE BROOK-ELKHORN 138.0 kV	DAM HTS-BABINE LAKE 138.0 kV	156.2	213	12.5
COLORADO-YAHARA RIV 138.0 kV	SNE - SANSUM N ELEC-MULTITRADE OF PITTSYLVANIA LP 138.0 kV	153.6	449	34.1
CHURCH-WELLSVILLE 138.0 kV line	YAHARA RIVER-W PT MUNICIPAL 138.0 kV	148.4	236	20.7
MCKENNA-ELKHORN 138.0 kV	DAM HTS-BABINE LAKE 138.0 kV	148.0	232	18.5
CEDAR GROVE-MULTITRADE OF PITTSYLVANIA LP 138.0 kV	NORTHEASTERN-MULTITRADE OF PITTSYLVANIA LP 138.0 kV	147.6	404	11.2
CHURCH-JACKSON RD 138.0 kV line	YAHARA RIV-W PT MUNICIPAL 138.0 kV Line	144.7	222	20.7
HUGHES LANE-S TACMA 138.0 kV	PACIFIC POWER-PERRNE 138.0 kV Line	143.2	335	5.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
VEVE CALZADA 2002-CEDAR GROVE 138.0 kV	MULTITRADE OF PITTSYLVANIA LP-NORTHEASTERN 138.0 kV	142.8	447	11.2
JACKSON RD-PASSAIC VALLEY WATER COMMISSION 138.0 kV L	YAHARA RIV-WEST POINT MUNICIPAL 138.0 kV	141.8	210	20.7
PASSAIC VALLEY WATER COMMISSION-YAHARA RIV 138.0 kV	YAHARA RIV-W PT MUNICIPAL 138.0 kV	140.8	216	43.4
MCKENNA SUBSTATION-NIETOS 138.0 kV L	DAM HTS-BABINE LAKE 138.0 kV	140.7	226	18.5
LANE-PASSAIC VLY WATER COMMISSION 138.0 kV L	COLORADO-YAHARA RIV 138.0 kV L	140.5	231	30.0
MONTVILLE-BEELNE 138.0 kV	AUDOBON 138 kV/1 kV Auto	136.6	278	16.1
JOHNNYCAKE-VEVE CALZADA 2002 138.0 kV	CEDAR GROVE-VEVE CALZADA 2002 SUBSTATION 138.0 kV	136.6	351	6.7
STANDARD OIL 3-SPADRA 138.0 kV	MULTITRADE OF PITTSYLVANIA P-SNE - SANSUM N ELEC 138.0 kV line	136.2	239	16.9
LAUREL AVE-VEVE CALZADA 2002 138.0 kV Cir. 2	LAUREL AVE-VEVE CALZADA 2002 138.0 kV Cir. 1	135.5	467	8.6
BABINE LAKE-DAM HTS 138.0 kV	PINE TREE ACRES-CRAWFORDSVILLE POWER PLANT 138.0 kV	135.4	577	5.4
WELLSVILLE-FORT SAGE 138.0 kV	WEST PT MUNICIPAL-YAHARA RIVER 138.0 kV Line	135.0	275	20.7
MONTVILLE-FT SAGE SUBSTATION 138.0 kV	AUDOBON 138 kV/1 kV Transformer	133.6	274	16.1
ALCHEM SUBSTATION-S TACOMA 138.0 kV L	PERRINE-PACIFIC POWER 138.0 kV	132.0	400	5.9
MULTITRADE OF PITTSYLVANIA LP-NORTHEASTERN 138.0 kV	CEDAR GROVE-VEVE CALZADA 2002 138.0 kV Line	130.7	483	11.8
WEST PT MUNICIPAL-OXFORD 138.0 kV L	FT SAGE-WELLSVILLE 138.0 kV	130.4	354	22.7
HARTS ISLAND SUBSTATION-BLACK HOLLOW 138.0 kV	YAHARA RIV-COLORADO 138.0 kV	129.9	215	6.9
ANTLER-GAYLORD CONTAINER BOGALUSA 138.0 kV	COLORADO-YAHARA RIV 138.0 kV	128.5	223	5.5
NORTHEASTERN SUBSTATION-JOHNNYCAKE 138.0 kV line	VEVE CALZADA 2002-CEDAR GROVE 138.0 kV	127.9	345	6.7
PIPE-LINDEN COGEN PLANT 138.0 kV	CAMP EVERS-LINDEN COGEN PLANT 138.0 kV	127.3	393	5.9
ANTLER-CREMATORIO 1512 138.0 kV	COLORADO SUBSTATION-YAHARA RIVER 138.0 kV	124.8	208	5.5
LAUREL AVE-VEVE CALZADA 2002 138.0 kV	VEVE CALZADA 2002-LAUREL AVE 138.0 kV	124.8	507	8.6
SADDLE BROOK-SHELDON RD 138.0 kV	BABINE LAKE-DAM HTS 138.0 kV	124.7	234	12.5
CAMP EVERS SUBSTATION-LINDEN COGEN PLANT SUBSTATION 138.0 kV	PIPE-LINDEN COGEN PLANT 138.0 kV	123.4	430	6.1
OAKES-LANE 138.0 kV line	YAHARA RIV-COLORADO 138.0 kV	122.8	211	30.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LEE VINING-LOCH MIRROR 161.0 kV	WELLS RIVER-GASTONIA PRIME POWER PARK 161.0 kV L	122.6	397	8.4
PERRINE-PACIFIC POWER 138.0 kV line	DAM HTS-BABINE LAKE 138.0 kV L	121.7	361	10.2
MAGNOLIA 1708-WHITE ROCK-NIKE 138.0 kV Line	PINE TREE ACRES-CRAWFORDSVILLE POWER PLANT 138.0 kV	121.3	668	8.8
CLOSTER-LEE VINING 161.0 kV	WELLS RIV-GASTONIA PRIME POWER PARK 161.0 kV	120.4	425	8.4
ARTESIAN-DAM HTS 138.0 kV line	CRAWFORDSVILLE POWER PLANT-PINE TREE ACRES 138.0 kV	120.0	302	5.9
JAN PHYL SUBSTATION-BABINE LAKE 138.0 kV	CRAWFORDSVILLE POWER PLANT-PINE TREE ACRES 138.0 kV	119.8	591	15.3
WEST CALDWELL-NORTHEASTERN 138.0 kV L	LAUREL AVE-ORANGE 345.0 kV	117.3	251	5.7
SNE - SANSUM N ELEC-JAN PHYL 138.0 kV Line	CRAWFORDSVILLE POWER PLANT-PINE TREE ACRES 138.0 kV	116.9	662	15.3
MAYWOOD-HARTS ISLAND 138.0 kV	COLORADO-YAHARA RIV SUBSTATION 138.0 kV	109.6	229	6.9
GAYLORD CONTAINER BOGALUSA-LOCH MIRROR 138.0 kV line	COLORADO-YAHARA RIV 138.0 kV	107.9	310	5.5
STANDARD OIL 3-BABINE LAKE 138.0 kV	MULTITRADE OF PITSYLVANIA LP-SNE - SANSUM N ELEC 138.0 kV	107.8	255	16.9
SHELDON ROAD-PACIFIC POWER 138.0 kV Line	DAM HTS-BABINE LAKE 138.0 kV	104.7	246	12.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NIETOS-COLORADO 138.0 kV L	Rebuild line with higher-capacity conductor	\$656,000	44% (\$292,000)
OLETA SUBSTATION-MONTCLAIR COGEN FACILITY 138.0 kV	Rebuild line with higher-capacity conductor	\$1,766,000	43% (\$767,000)
WELLS RIV SUBSTATION-GASTONIA PRIME POWER PARK 161.0 kV	Rebuild line with higher-capacity conductor	\$823,000	25% (\$205,000)
SPADRA SUBSTATION-MONTCLAIR COGEN FACILITY 138.0 kV	Rebuild line with higher-capacity conductor	\$1,161,000	43% (\$504,000)
OLETA-COLORADO 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,105,000	43% (\$914,000)
ATHENIA-DAM HTS 138.0 kV	Rebuild line with higher-capacity conductor	\$2,349,000	28% (\$646,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SADDLE BROOK-ELKHORN 138.0 kV	Rebuild line with higher-capacity conductor	\$423,000	30% (\$127,000)
COLORADO-YAHARA RIV 138.0 kV	Rebuild line with higher-capacity conductor	\$1,024,000	45% (\$462,000)
CHURCH-WELLSVILLE 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,218,000	60% (\$724,000)
ELKHORN-MCKENNA 138.0 kV L	Rebuild line with higher-capacity conductor	\$509,000	44% (\$226,000)
CEDAR GROVE-MULTITRADE OF PITTSYLVANIA LP 138.0 kV	Rebuild line with higher-capacity conductor	\$2,476,000	26% (\$633,000)
CHURCH-JACKSON RD 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,851,000	60% (\$2,289,000)
HUGHES LANE-S TACMA 138.0 kV	Rebuild line with higher-capacity conductor	\$1,053,000	8% (\$85,000)
VEVE CALZADA 2002-CEDAR GROVE 138.0 kV	Rebuild line with higher-capacity conductor	\$5,548,000	26% (\$1,418,000)
JACKSON RD-PASSAIC VALLEY WATER COMMISSION 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,494,000	60% (\$888,000)
PASSAIC VALLEY WATER COMMISSION-YAHARA RIV 138.0 kV	Rebuild line with higher-capacity conductor	\$1,226,000	100% (\$1,226,000)
MCKENNA SUBSTATION-NIETOS 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,725,000	44% (\$1,213,000)
LANE-PASSAIC VLY WATER COMMISSION 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,754,000	61% (\$2,283,000)
BEELINE-MONTVILLE 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,066,000	53% (\$2,666,000)
VEVE CALZADA 2002-JOHNNYCAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,956,000	22% (\$660,000)
STANDARD OIL 3-SPADRA 138.0 kV	Rebuild line with higher-capacity conductor	\$1,890,000	43% (\$820,000)
LAUREL AVE-VEVE CALZADA2002 138.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,109,000	23% (\$482,000)
BABINE LAKE-DAM HTS 138.0 kV	Rebuild line with higher-capacity conductor	\$901,000	8% (\$70,000)
WELLSVILLE-FORT SAGE 138.0 kV	Rebuild line with higher-capacity conductor	\$6,488,000	60% (\$3,857,000)
MONTVILLE-FT SAGE SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,179,000	53% (\$1,147,000)
ALCHEM SUBSTATION-S TACOMA 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,214,000	8% (\$258,000)
MULTITRADE OF PITTSYLVANIA LP-NORTHEASTERN 138.0 kV	Rebuild line with higher-capacity conductor	\$7,779,000	26% (\$2,055,000)
OXFORD-WEST PT MUNICIPAL 138.0 kV	Rebuild line with higher-capacity conductor	\$10,363,000	55% (\$5,704,000)
HARTS ISLAND SUBSTATION-BLACK HOLLOW 138.0 kV	Rebuild line with higher-capacity conductor	\$2,251,000	22% (\$506,000)
ANTLER-GAYLORD CONTAINER BOGALUSA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,847,000	75% (\$2,138,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JOHNNYCAKE-NORTHEASTERN SUBSTATION 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,408,000	22% (\$538,000)
PIPE-LINDEN COGEN PLANT 138.0 kV	Rebuild line with higher-capacity conductor	\$1,547,000	11% (\$176,000)
ANTLER-CREMATORIO 1512 138.0 kV	Rebuild line with higher-capacity conductor	\$2,421,000	75% (\$1,818,000)
LAUREL AVE-VEVE CALZADA 202 138.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,915,000	23% (\$437,000)
SHELDON ROAD-SADDLE BROOK 138.0 kV	Rebuild line with higher-capacity conductor	\$2,069,000	30% (\$620,000)
CAMP EVERS-LINDEN COGEN PLANT 138.0 kV	Rebuild line with higher-capacity conductor	\$693,000	11% (\$75,000)
OAKES-LANE 138.0 kV line	Rebuild line with higher-capacity conductor	\$10,361,000	61% (\$6,302,000)
LEE VINING-LOCH MIRROR 161.0 kV	Rebuild line with higher-capacity conductor	\$4,293,000	30% (\$1,285,000)
PERRINE-PACIFIC POWER 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,347,000	30% (\$399,000)
MAGNOLIA 1708-WHITE ROCK-NIKE 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,296,000	6% (\$73,000)
CLOSTER-LEE VINING 161.0 kV	Rebuild line with higher-capacity conductor	\$3,080,000	30% (\$921,000)
ARTESIAN-DAM HTS 138.0 kV	Rebuild line with higher-capacity conductor	\$1,878,000	11% (\$215,000)
BABIN LAKE-JAN PHYL 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,971,000	23% (\$458,000)
NORTHEASTERN-W CALDWELL 138.0 kV	Rebuild line with higher-capacity conductor	\$5,258,000	40% (\$2,076,000)
SNE - SANSUM N ELEC-JAN PHYL 138.0 kV Line	Rebuild line with higher-capacity conductor	\$998,000	23% (\$232,000)
MAYWOOD-HARTS ISLAND 138.0 kV	Reconductor line	\$558,000	22% (\$125,000)
GAYLORD CONTAINER BOGALUSA-LOCH MIRROR 138.0 kV line	Reconductor line	\$727,000	75% (\$546,000)
STANDARD OIL 3-BABINE LAKE 138.0 kV	Reconductor line	\$186,000	43% (\$81,000)
SHELDON RD-PACIFIC POWER 138.0 kV Line	Reconductor line	\$397,000	30% (\$119,000)
	Total allocated to project		\$51,761,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.